Supervised method

Absolute Frame Error / Offset Deviation

- Explanation: This is the most direct basic metric. It simply calculates the absolute difference between the frame offset predicted by your algorithm and the actual known ground truth offset. It directly tells you by how many frames or milliseconds your algorithm was incorrect

- Source: Beyond Audio and Pose: A General-Purpose Framework for Video Synchronization and Camera Synchronization from Feature Points for 3D Motion Reconstruction

Temporal RMSE (Root Mean Square Error)

- Explanation: Calculates the square root of the average of the squared differences between the predicted matching time and the ideal reference time. Unlike a simple average, RMSE heavily penalizes large isolated errors (outliers). It is the perfect metric for measuring clock drift or jitter over time

- Source: Evaluation Frameworks for Spatiotemporal Video Alignment in Multi-Traversal Railway Inspection

TCOM & RPOM (Railway Spatial Error Metrics)

- Explanation: Specific to your railway use case! TCOM (Track Centerline Offset Metric) measures the lateral deviation in millimeters between the calculated track centerline and the ground truth. RPOM (Rail Position Offset Metric) does the same for the outer edges of the rails. These metrics are calculated by projecting the image into a Bird's Eye View (BEV) to evaluate the absolute accuracy of the spatial overlay

- Source: A COMPREHENSIVE FRAMEWORK FOR EVALUATING VISION-BASED ON-BOARD RAIL TRACK DETECTION

Unsupervised method

Average content distance (ACD)

- Explanation: This metric measures the average pixel-level distance between consecutive frames in your aligned video sequence. A very low score ensures that the transition from one video to the other is smooth and free of "flickering", confirming a good alignment without needing ground truth data

- Source : Evaluation Frameworks for Spatiotemporal Video Alignment in Multi-Traversal Railway Inspection

Flicker penalty (FP)

- Explanation: This method isolates high-frequency "temporal noise" (pixels that change abruptly and illogically). It uses "optical flow" to mathematically predict the next frame from the current one, and then compares it with the actual next frame in the aligned sequence. If they match (error close to 0), the video is well-synchronized and visually stable

- Source: World Consistency Score: A Unified Metric for Video Generation Quality - arXiv

World consistency score (WCS)

- Explanation: This is a global metric that verifies if the aligned video respects real-world logic and physics. It combines four sub-metrics: object permanence (an object does not suddenly disappear), relation stability, causal compliance, and flicker penalty (FP). This autonomously quantifies the temporal and physical coherence of a video scene.

- Source: World Consistency Score: A Unified Metric for Video Generation Quality - arXiv

Fréchet Video Distance (FVD) & Temporal LPIPS

- Explanation: These are perceptual metrics driven by neural networks. FVD evaluates global temporal drift by comparing the statistical distribution of your aligned video with that of natural, smooth video sequences. Temporal LPIPS evaluates subtle, unnatural appearance changes and identity drift between consecutive frames

- Source: Temporal Drift in AI-Generated Video: Causes, Evaluation, and Production Strategies - iMerit and Evaluation Frameworks for Spatiotemporal Video Alignment in Multi-Traversal Railway Inspection

Photometric Comparison with Min/Max Filters

- Explanation: To evaluate if two warped frames overlap well without being tricked by micro-movements or exposure changes, this method uses a min/max pixel comparison. Instead of a strict pixel-by-pixel subtraction, it compares each pixel of the primary video to the minimum and maximum values of a small neighborhood in the secondary video, providing a highly robust matching score

- Source: Video Matching (Robotics, Vision, and Sensor Networks Group)

Personnal notes

In my opinion, an effective way to evaluate video alignment would be to measure the quantity and quality of the feature matches. Interestingly, papers focusing on global video alignment do not use this as a final evaluation metric, preferring macro-metrics like temporal error or visual coherence. However, the quantity and quality of links are heavily utilized as internal scoring mechanisms (such as the GMS algorithm or Lowe's ratio test) to filter and validate correspondences during the alignment process. Furthermore, they actually serve as the standard evaluation metrics—measured as Precision, Recall, and Inlier Ratio—when researchers evaluate the performance of the feature matching algorithms themselves.